Cessna 152 (C152)

Lycoming O-235-L2C, 110 hp · 2 seats · Typical 152 (1978–1985) POH procedures; carburetted engine, 30° flaps, gravity-fed fuel with shutoff valve.

V-speeds & limits

Vr	**50 KIAS**
Vx	**55 KIAS**
Vy	**67 KIAS**
Best glide	**60 KIAS**
Vs / Vso	**40 / 35 KIAS**
Vfe	**85 KIAS**
Va	**104 KIAS (1,670 lb) / 93 KIAS (1,500 lb)**
Vno	**111 KIAS**
Vne	**149 KIAS**
Approach	**60–70 KIAS flaps up / 55–65 KIAS flaps 30°**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Fuel quantity	**CHECK visually (24.5 gal usable std)**
Oil quantity	**CHECK (6 qt max, 4 qt min)**
Passenger briefing	**COMPLETE (belts, doors, exits, sterile cockpit)**
Seats, belts, harnesses	**ADJUST & LOCK**
Brakes	**TEST & SET**
Control lock	**REMOVE**
Fuel shutoff valve	**ON**
Avionics power / radios	**OFF**
Electrical equipment	**OFF**
Circuit breakers	**CHECK IN**
Carburettor heat	**COLD (push in)**

Engine Start

Mixture	**RICH**
Primer	**AS REQUIRED (1–3 strokes, then IN & LOCKED)**
Throttle	**OPEN 1/2 inch**
Master switch	**ON**
Beacon	**ON**
Propeller area	**CLEAR**
Ignition switch	**START (release when engine fires)**
Throttle	**ADJUST 1,000 RPM**
Oil pressure	**CHECK rising within 30 s (60 s cold)**
Ammeter	**CHECK charging**
Avionics power / radios	**ON**
Flashing beacon / nav lights	**AS REQUIRED**

Before Taxi

Altimeter	**SET**
Heading indicator	**SET to compass**
Radios / transponder	**SET**
Flaps	**CHECK operation, then UP**
Parking brake	**RELEASE**
Brakes	**CHECK during taxi**
Flight instruments	**CHECK during turns (DG, turn coordinator, compass)**

Before Takeoff

Parking brake	**SET**
Seats, belts, harnesses	**CHECK SECURE**
Cabin doors	**CLOSED & LATCHED**
Flight controls	**FREE & CORRECT**
Flight instruments	**SET**
Fuel shutoff valve	**ON**
Mixture	**RICH (below 3,000 ft)**
Elevator trim	**TAKEOFF**
Throttle	**1,700 RPM**
Magnetos	**CHECK (max drop 125 RPM, max 50 RPM differential)**
Carburettor heat	**CHECK (RPM drop, then COLD)**
Engine instruments / ammeter	**CHECK (oil press 60–90 psi green)**
Suction gauge	**CHECK (4.5–5.4 in Hg)**
Throttle	**1,000 RPM**
Throttle friction lock	**ADJUST**
Radios / transponder	**SET / ALT**
Strobe lights	**ON**
Wing flaps	**0°–10°**
Takeoff briefing	**COMPLETE**

After Takeoff / Climb

Airspeed	**67 KIAS (Vy) / 55 KIAS (Vx, obstacles)**
Wing flaps	**UP (above 60 KIAS)**
Throttle	**FULL OPEN**
Mixture	**RICH (lean above 3,000 ft for smooth operation)**
Engine instruments	**CHECK**
Landing light	**OFF**

Cruise

Power	**2,000–2,550 RPM (75% max continuous)**
Elevator trim	**ADJUST**
Mixture	**LEAN as required**
Engine instruments	**CHECK**
Fuel quantity / endurance	**CHECK**
Heading indicator	**ALIGN with compass**

Descent

Altimeter	**SET**
Mixture	**RICH (adjust for smooth operation)**
Power	**AS DESIRED**
Carburettor heat	**AS REQUIRED (full heat before closing throttle)**
Approach briefing	**COMPLETE**

Before Landing

Seats, belts, harnesses	**SECURE**
Fuel shutoff valve	**ON**
Mixture	**RICH**
Carburettor heat	**ON (full)**
Airspeed	**60–70 KIAS (flaps up)**
Wing flaps	**AS DESIRED (0°–10° below 110 KIAS, 10°–30° below 85 KIAS)**
Airspeed	**55–65 KIAS (flaps down)**
Landing light	**ON**

After Landing

Wing flaps	**UP**
Carburettor heat	**COLD**
Strobe lights / landing light	**OFF / AS REQUIRED**
Transponder	**STBY / as required**

Shutdown

Parking brake	**SET**
Throttle	**IDLE (1,000 RPM then idle)**
Avionics power / radios	**OFF**
Electrical equipment	**OFF**
Mixture	**IDLE CUTOFF (pull full out)**
Ignition switch	**OFF (key removed)**
Master switch	**OFF**
Control lock	**INSTALL**
Fuel shutoff valve	**ON (leave on; OFF only for extended parking)**

Abnormal procedures

Alternator Failure / Ammeter Discharge

Ammeter shows discharge or low-voltage light on

Avionics power	**OFF**
Master switch	**OFF (both sides)**
Master switch	**ON**
Low-voltage light / ammeter	**CHECK**
Avionics power	**ON**

▶ Charging restored?

Yes:

Ammeter	**MONITOR**

No:

Alternator (master ALT half)	**OFF**
Non-essential electrical equipment	**OFF**
Battery	**CONSERVE for flaps and radio on approach**
Deferred: Alternator failed – on battery only, limited endurance. Land as soon as practical, lower flaps early while power remains.	**NOTED**

Low Oil Pressure

Oil pressure below green arc (60 psi) in flight

Oil pressure / oil temperature	**CHECK**

▶ Oil temperature also rising?

Yes:

Power	**REDUCE to minimum required**
Engine failure	**EXPECT – plan a precautionary landing**
Deferred: Low oil pressure with high temperature – land as soon as possible, expect engine failure.	**NOTED**

No – temp normal:

Flight	**CONTINUE to nearest suitable airport, monitor**
Deferred: Low oil pressure indication – land as soon as practical, maintenance check.	**NOTED**

High Oil Temperature

Mixture	**RICH**
Power	**REDUCE**
Airspeed	**INCREASE (lower nose for cooling)**
Oil pressure	**MONITOR**

▶ Temperature returning to green arc?

Yes:

Flight	**CONTINUE, monitor**

No:

Deferred: High oil temperature – land as soon as practical, minimum power.	**NOTED**

Rough Running / Carburettor Ice

Unexplained RPM drop or rough running

Carburettor heat ..... **ON (full)**

Expect an initial further RPM drop and roughness as ice melts; leave heat on.

Mixture ..... **ADJUST for smooth operation**

▶ RPM recovered / running smooth?

Yes:

Carburettor heat ..... **AS REQUIRED (consider partial/full for remainder of flight)**

No:

Magnetos ..... **CHECK L / R / BOTH, select smoothest**

Primer ..... **CHECK IN & LOCKED**

Fuel shutoff valve ..... **ON**

Deferred: Rough running engine – land as soon as practical, be ready for engine failure. ..... **NOTED**

Fuel Starvation / Low Fuel

Fuel shutoff valve ..... **ON**

Fuel quantity ..... **CHECK both gauges**

Mixture ..... **RICH**

Attitude ..... **WINGS LEVEL (uncoordinated flight unports a tank)**

Deferred: Low fuel – land at nearest suitable airport, keep wings level on approach. ..... **NOTED**

Flap Failure

Flap circuit breaker ..... **CHECK / RESET once**

Flap switch ..... **CYCLE**

▶ Flaps operable?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Approach ..... **FLAPS UP – 60–70 KIAS, flatter approach, longer runway**

Deferred: Flaps inoperative – flaps-up landing at 60–70 KIAS, increased landing distance. ..... **NOTED**

Door Open in Flight

Airspeed ..... **REDUCE (approx 65–70 KIAS)**

Aircraft ..... **CONTROL – door has little effect on handling**

Window on open-door side ..... **OPEN (equalise pressure), then door CLOSE & LATCH**

▶ Door latched?

Yes:

Flight ..... **CONTINUE**

No:

Deferred: Door unlatched – land as soon as practical, normal landing, keep speed low. ..... **NOTED**

Vacuum Failure

Suction gauge below 4.5 in Hg or gyro flags

Suction gauge ..... **CHECK**

Attitude indicator / heading indicator ..... **DISREGARD – cross-check**

Turn coordinator, compass, altimeter, airspeed ..... **USE for partial-panel flight**

IMC ..... **AVOID / EXIT**

Deferred: Vacuum failed – attitude and heading indicators unreliable; VMC only. ..... **NOTED**

Static Source Blocked

Erroneous airspeed / altimeter / VSI

Alternate static source ..... **PULL ON (if installed)**

Cabin vents / windows ..... **CLOSED when alternate static in use**

Airspeed / altimeter ..... **APPLY POH alternate static corrections**

▶ Alternate static installed?

Yes:

Deferred: Alternate static in use – airspeed and altimeter read slightly high; add margin on approach. ..... **NOTED**

No:

VSI glass ..... **BREAK as last resort (reverse-reading VSI)**

Deferred: Static blocked – fly attitude and power settings; airspeed indications unreliable. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

On the runway

Throttle (memory)	IDLE
Brakes (memory)	APPLY
Wing flaps (memory)	RETRACT
Mixture	IDLE CUTOFF
Ignition switch	OFF
Master switch	OFF

Engine Failure After Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Airspeed (memory)	60 KIAS
Mixture (memory)	IDLE CUTOFF
Fuel shutoff valve (memory)	OFF
Ignition switch (memory)	OFF
Wing flaps (memory)	AS REQUIRED
Master switch (memory)	OFF

Land straight ahead with only small changes of direction. Do not attempt to turn back to the runway.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	60 KIAS (best glide)
Carburettor heat (memory)	ON
Primer (memory)	IN & LOCKED
Fuel shutoff valve (memory)	ON
Mixture (memory)	RICH
Ignition switch (memory)	BOTH (START if propeller stopped)

▶ Engine restarted?

Yes:

Power	AS REQUIRED
Engine instruments	MONITOR
Carburettor heat	AS REQUIRED
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect carb ice or fuel starvation.	NOTED

No – forced landing:

Forced landing without engine power:

Field / wind	SELECT & PLAN (into wind if possible)
Mayday / 7700	TRANSMIT (time permitting)
Seats, belts, harnesses	SECURE
Airspeed	65 KIAS flaps up / 60 KIAS flaps down
Mixture	IDLE CUTOFF
Fuel shutoff valve	OFF
Ignition switch	OFF
Wing flaps	AS REQUIRED (30° recommended)
Master switch	OFF (when landing assured)
Doors	UNLATCH prior to touchdown
Touchdown	SLIGHTLY TAIL LOW
Brakes	APPLY HEAVILY

Engine Fire During Start

Cranking (memory) ..... **CONTINUE (draw flames into engine)**

▶ Engine started?

Yes:

Power ..... **1,700 RPM for a few minutes**

Engine ..... **SHUT DOWN & INSPECT for damage**

No:

Throttle ..... **FULL OPEN**

Mixture ..... **IDLE CUTOFF**

Cranking ..... **CONTINUE**

Fuel shutoff valve ..... **OFF**

Ignition switch ..... **OFF**

Master switch ..... **OFF**

Fire extinguisher ..... **OBTAIN & USE**

Aircraft ..... **EVACUATE, inspect for damage**

Engine Fire in Flight

LAND IMMEDIATELY

Mixture (memory) ..... **IDLE CUTOFF**

Fuel shutoff valve (memory) ..... **OFF**

Master switch (memory) ..... **OFF**

Cabin heat / air (memory) ..... **OFF (except overhead vents)**

Airspeed (memory) ..... **85 KIAS (increase if fire persists)**

Do not attempt an engine restart.

Forced landing ..... **EXECUTE (see Engine Failure in Flight)**

Electrical Fire in Flight

LAND AS SOON AS POSSIBLE

Master switch (memory) ..... **OFF**

Avionics power / radios (memory) ..... **OFF**

All other switches (except ignition) (memory) ..... **OFF**

Vents / cabin air / heat ..... **CLOSED**

Fire extinguisher ..... **ACTIVATE (if available)**

After discharging the extinguisher, ventilate the cabin.

▶ Fire out and electrical power necessary?

Yes:

Master switch ..... **ON**

Circuit breakers ..... **CHECK – do not reset a faulted breaker**

Radios / electrical ..... **ON one at a time, with delay, check for smoke**

Vents / cabin air / heat ..... **OPEN when fire is completely out**

Deferred: Electrical fire – faulted circuit isolated, affected equipment unavailable. ..... **NOTED**

No:

Continue with master off. No flaps, radios, lights or fuel gauges available.

Deferred: Electrical power off – manual (no-flap) landing, no radio; squawk by light signals. ..... **NOTED**

Cabin Fire

LAND AS SOON AS POSSIBLE

Master switch (memory) ..... **OFF**

Vents / cabin air / heat (memory) ..... **CLOSED**

Fire extinguisher ..... **ACTIVATE**

Cabin ..... **VENTILATE after fire is out**

Land ..... **AS SOON AS POSSIBLE, inspect for damage**

Emergency Descent

Throttle (memory)	**IDLE**
Carburettor heat (memory)	**ON**
Wing flaps (memory)	**UP**
Airspeed (memory)	**111 KIAS smooth air / 104 KIAS turbulence (Va)**
Mixture	**RICH**
Power	**CLEAR ENGINE periodically**

Spin Recovery

Inadvertent spin

Ailerons (memory)	**NEUTRAL**
Throttle (memory)	**IDLE**
Rudder (memory)	**FULL OPPOSITE to direction of rotation**
Control wheel (memory)	**BRISKLY FORWARD (just after rudder reaches stop)**
Controls (memory)	**HOLD until rotation stops**
Rudder (memory)	**NEUTRALISE as rotation stops**
Recovery	**SMOOTH pull-out from the dive**

Flaps must be retracted; if extended, retract immediately after recovery. Premature relaxation of controls may prolong the spin.

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.8 (build c9e52546) · 2026-11-07